Настройка сетевых сервисов

DNS и DHCP в Cisco Packet Tracer

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Цели и задачи работы

Цель лабораторной работы

Изучить принципы работы протоколов DNS и DHCP
и получить практические навыки их настройки
в среде Cisco Packet Tracer.

Настройка DNS-сервера

Добавление сервера в сеть

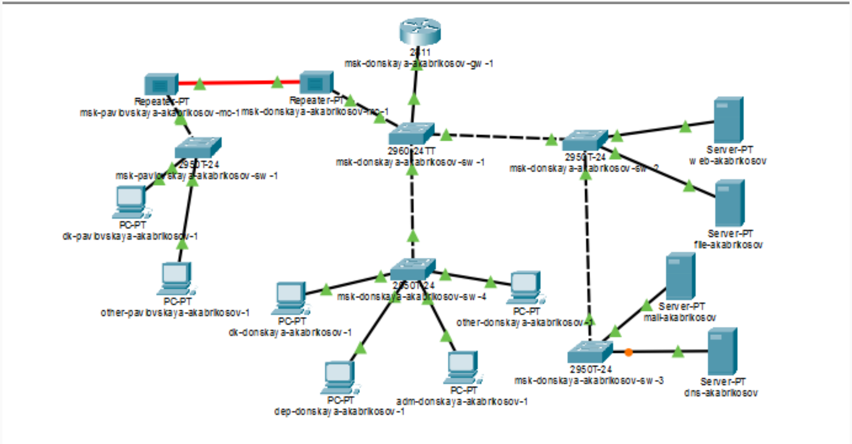

Рис. 1: Топология сети

Настройка IP-адреса сервера

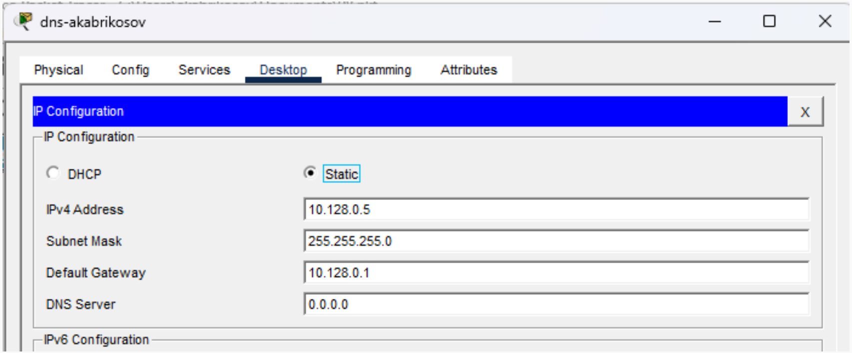

Рис. 2: IP-конфигурация DNS

Настройка DNS-записей

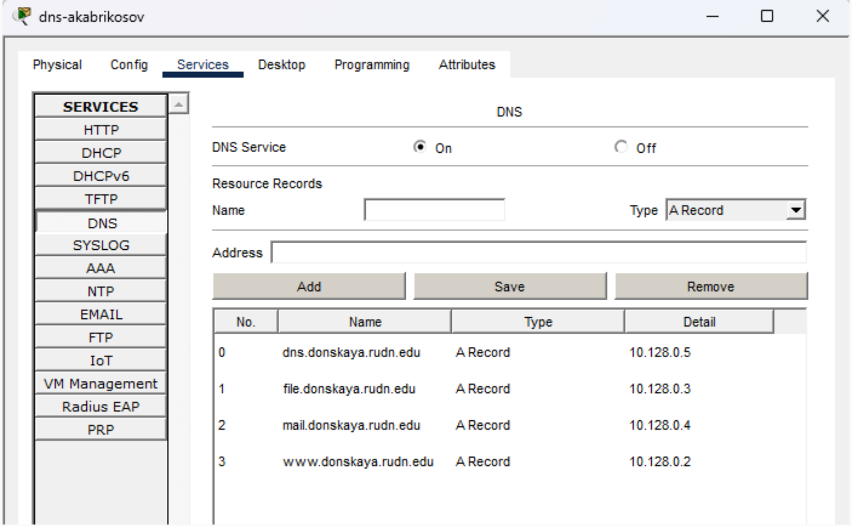

Рис. 3: DNS записи

Настройка DHCP

Конфигурация DHCP на маршрутизаторе

```
msk-donskaya-akabrikosov-gw-1>enab
Password:
msk-donskaya-akabrikosov-gw-1#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-akabrikosov-gw-1(config)#
msk-donskaya-akabrikosov-gw-1(config)#ip name-server 10.128.0.5
msk-donskaya-akabrikosov-gw-1(config)#service dhcp
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp pool dk
msk-donskaya-akabrikosov-gw-1(dhcp-config)#network 10.128.3.0 255.255.255.0
msk-donskaya-akabrikosov-gw-1(dhcp-config)#default-router 10.128.3.1
msk-donskaya-akabrikosov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-akabrikosov-gw-1(dhcp-config)#exit
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.3.1 10.128.3.29
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.3.200 10.128.3.254
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp pool departaments
msk-donskaya-akabrikosov-gw-1(dhcp-config)#network 10.128.4.0 255.255.255.0
msk-donskaya-akabrikosov-gw-1(dhcp-config)#default-router 10.128.4.1
msk-donskaya-akabrikosov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-akabrikosov-gw-1(dhcp-config)#ex
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.4.1 10.128.4.29
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.4.200 10.128.4.254
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp pool adm
msk-donskaya-akabrikosov-gw-1(dhcp-config)#network 10.128.5.0 255.255.255.0
msk-donskaya-akabrikosov-gw-1(dhcp-config)#default-router 10.128.5.1
msk-donskaya-akabrikosov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-akabrikosov-gw-1(dhcp-config)#ex
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.5.1 10.128.5.29
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.5.200 10.128.5.254
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp pool other
msk-donskaya-akabrikosov-gw-1(dhcp-config)#network 10.128.6.0 255.255.255.0
msk-donskaya-akabrikosov-gw-1(dhcp-config)#default-router 10.128.6.1
msk-donskaya-akabrikosov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-akabrikosov-gw-1(dhcp-config)#ex
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.6.1 10.128.6.29
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.6.200 10.128.6.254
msk-donskaya-akabrikosov-gw-1(config)#ex
msk-donskaya-akabrikosov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

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Пулы DHCP

- dk — 10.128.3.0/24
- departments — 10.128.4.0/24
- adm — 10.128.5.0/24
- other — 10.128.6.0/24

Работа клиентов

Получение адреса по DHCP

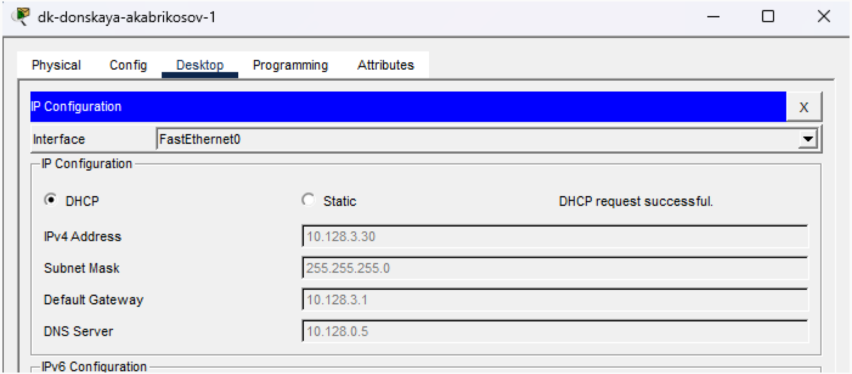

Рис. 5: DHCP на ПК

Информация о пулах

```
msk-donskaya-akabrikosov-gw-1#sh ip dhcp pool

Pool dk :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 8
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.128.3.1           10.128.3.1       - 10.128.3.254      2    / 8     / 254

Pool departaments :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 8
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.128.4.1           10.128.4.1       - 10.128.4.254      1    / 8     / 254

Pool adm :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 8
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
```

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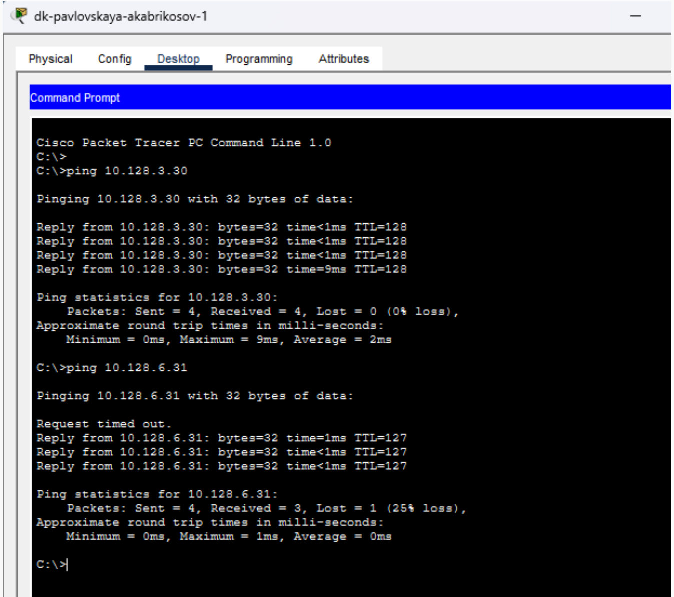
Таблица выданных адресов

```
msk-donskaya-akabrikosov-gw-1#
msk-donskaya-akabrikosov-gw-1#sh ip dhcp bin
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.128.3.30      00E0.B0E6.6B91           --                     Automatic
10.128.3.31      00D0.BC46.854E           --                     Automatic
10.128.4.30      0060.5CEA.6ED7           --                     Automatic
10.128.5.30      0001.43D6.CEA1           --                     Automatic
10.128.6.30      000B.BE06.7A0B           --                     Automatic
10.128.6.31      0090.2182.8188           --                     Automatic
msk-donskaya-akabrikosov-gw-1#
```

Рис. 7: DHCP binding

Проверка сети

Проверка доступности



Связь между подсетями

dep-donskaya-akabrikosov-1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.128.3.30

Pinging 10.128.3.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.3.30: bytes=32 time<1ms TTL=127
Reply from 10.128.3.30: bytes=32 time<1ms TTL=127
Reply from 10.128.3.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.5.30

Pinging 10.128.5.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.5.30: bytes=32 time<1ms TTL=127
Reply from 10.128.5.30: bytes=32 time<1ms TTL=127
Reply from 10.128.5.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.5.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.6.30

Pinging 10.128.6.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.6.30: bytes=32 time=1ms TTL=127
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127
Reply from 10.128.6.30: bytes=32 time=1ms TTL=127
```

Работа DHCP (Simulation)

Процесс DHCP

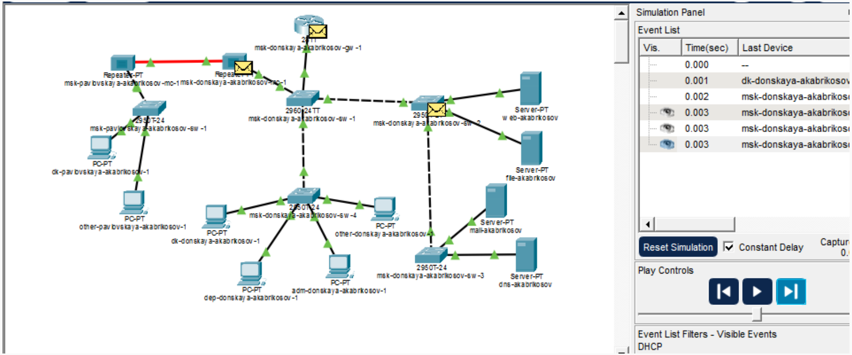


Рис. 10: DHCP Simulation

Discover

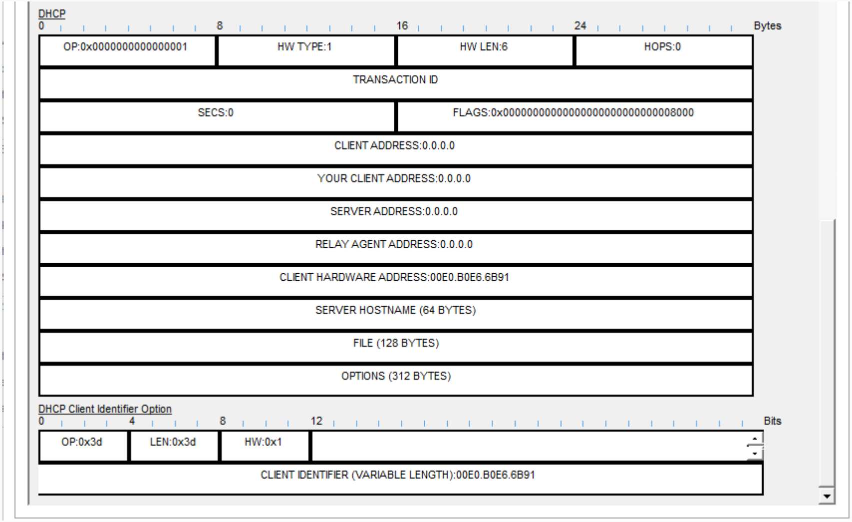
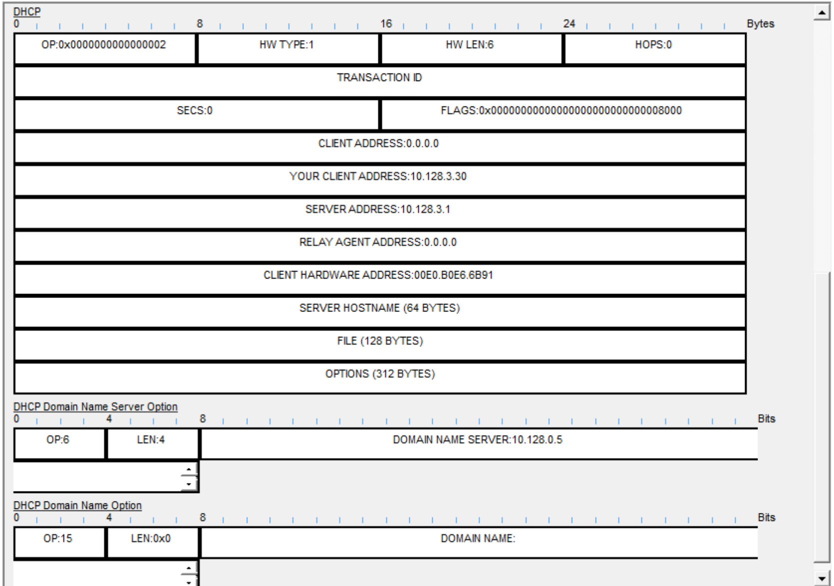


Рис. 11: DHCP Discover

Offer


DHCP
0
8
16
24
Bytes
OP:0x0000000000000002
HW TYPE:1
HW LEN:6
HOPS:0
TRANSACTION ID
SECS:0
FLAGS:0x00000000000000000000000000008000
CLIENT ADDRESS:0.0.0.0
YOUR CLIENT ADDRESS:10.128.3.30
SERVER ADDRESS:10.128.3.1
RELAY AGENT ADDRESS:0.0.0.0
CLIENT HARDWARE ADDRESS:00E0.B0E6.6B91
SERVER HOSTNAME (64 BYTES)
FILE (128 BYTES)
OPTIONS (312 BYTES)
DHCP Domain Name Server Option
0
4
8
Bits
OP:6
LEN:4
DOMAIN NAME SERVER:10.128.0.5
DHCP Domain Name Option
0
4
8
Bits
OP:15
LEN:0x0
DOMAIN NAME:

Request

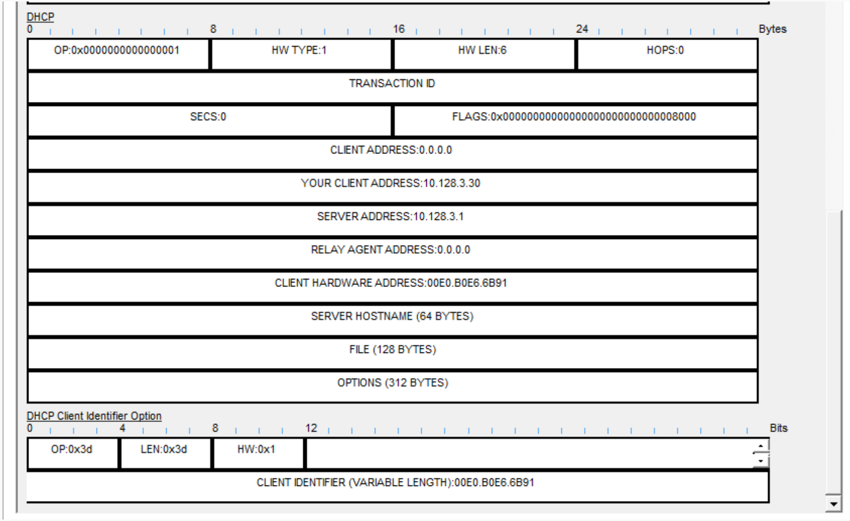


Рис. 13: DHCP Request

ACK

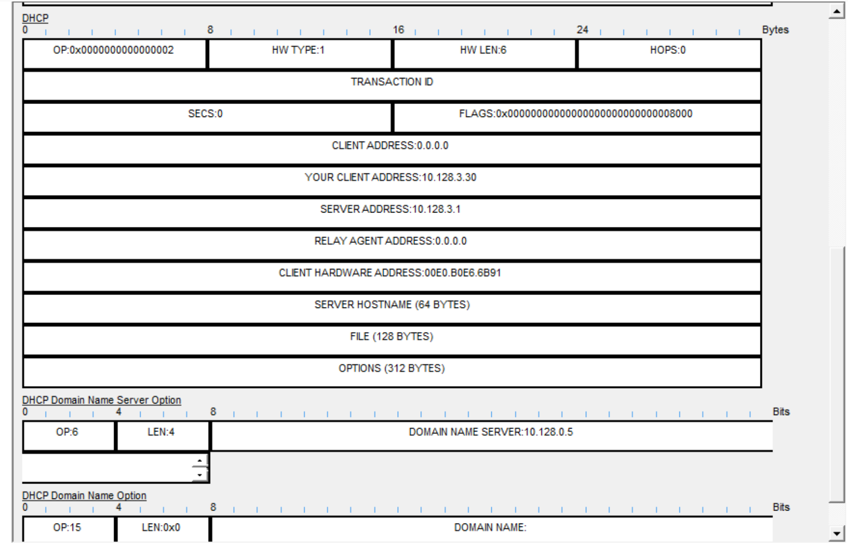


Рис. 14: DHCP ACK

Вывод

Итоги работы

Настроены сервисы DNS и DHCP в сети Cisco Packet Tracer.

Реализовано автоматическое распределение IP-адресов
и разрешение доменных имён.

Исследован процесс DHCP и структура сообщений.

Подтверждена работоспособность сети и взаимодействие подсетей.